

MyDesign Portfolio - Yung Gadi and the Dusty Bunch

Element E: Application of STEM Principles and Practices

Template Instructions This template is designed for technical writing and succinctness. Replace placeholders with precise details from your project, ensuring your responses are backed by evidence to fully meet the rubric's criteria.

E1. Application of STEM Principles to Design Requirements

- **Design Requirements Overview:**
 - How quick it opens: able to fully open and close in 5 sec: high priority
 - How far it opens: able to open in fully: High priority
 - Easily usable by students: give it to 3 students: High Priority
- **STEM Applications:**
 - **Example Application:** Stem principles apply to this project because the energy that we put into the lever to open the contraption helps the blast gate get open
 - **Calculations and Theories:** $E_1 = E_2$

E2. Substantiation of Proposed Solution with STEM

- **Solution Overview:** Use a lever to open the blast gate
- **STEM Justification:** Explain how STEM principles substantiate your solution.
 - **Detailed Substantiation:** The energy put into pushing the lever is the same amount that is used to open the blast gate.

E3. Expert Review of STEM Applications

- **Expert Consultation:** Expert consultations helped us come up with the idea and prototype
 - **Feedback Summary:** Feedback that influenced our project included making it more visible to the students so that they will remember to use the blast gate before turning on any dust-producing machine.

E4. Impact of Expert Review on Design

- **Verification and Corrective Responses:** expert review changed our whole idea of the project as we were originally going to make an automatic door closer to reduce the amount of dust getting out of the workshop.
 - **Changes Made:** Open and closed signs, using wood instead of 3D print, using bike wires
-